

SAILAKSHMI B T

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PROFESSIONAL SUMMARY

Robotics and Automation undergraduate (CGPA 9.10/10.0) with hands-on experience in embedded systems, robotics, and computer vision. Skilled in Arduino-based prototyping, sensor interfacing, and AI-driven applications. Seeking internship opportunities in robotics, embedded systems, or AI/computer vision.

EDUCATION

B.E., Robotics and Automation, Madras Institute of Technology (CGPA: 9.10/10.0) 2024 - 2028
Higher Secondary Certificate (2024): 94.16% | SSC (2022): 95.2% | JEE (Main) 2024: 81.82 percentile

EXPERIENCE

- Computer Vision and AI Intern — PixelBotix** June 2026
- Completed a 4-week intensive in image processing and object detection, building and deploying 3 computer vision and AI projects from concept to demo.
- Industrial Robotics and Automation Training — Birla Carbon and KUKA** Feb 2026
- Trained in industrial robotic arm programming, gaining hands-on exposure to Industry 4.0 concepts and KUKA safety protocols.

PROJECTS

- AgileX Scout Mini ROS 2 Jazzy Integration** ROS 2 Jazzy, C++, Python, Gazebo, RViz2
- Developed ROS 2 Jazzy driver nodes for real-time telemetry and odometry tracking, validated across multiple Gazebo/RViz2 simulation scenarios.
- Student Life Concierge AI Agent** Google ADK, Gemini API, FastAPI, Python, Pytest
- Built a multi-agent FastAPI backend with prompt-injection defenses to automate student scheduling, using Google ADK and Gemini API as part of the Kaggle x Google 5-Day AI Agents Intensive.
- Single-Axis Gimbal Stabilization System** Arduino Nano, MPU-6050, SG90 Servo
- Engineered a closed-loop PID gimbal stabilizer using an MPU-6050 IMU on Arduino Nano, achieving sub-degree stabilization accuracy.

DESIGN & PATENTS

- Mechanical Rotary Device** Design No. 476802-001 | SolidWorks, Design Patent Pending
- Co-Designed a dual-mode Geneva wheel and dog clutch rotary mechanism in SolidWorks, combining two motion-control functions into one assembly; filed for design patent.

HACKATHONS & COMPETITIONS

- Smart India Hackathon 2025 - Finalist** Delhi, 2025
- Collaborated on reverse-engineering multi-architecture firmware binaries to identify cryptographic primitives and protocols using AI and machine learning.
- H2TS Hackathon 2025 - 2nd Place** Dec 2025
- Worked with a cross-functional team to build and deploy an AI-driven SOS platform for rapid emergency response.

TECHNICAL SKILLS

Languages: Python (Intermediate), Embedded C
Control & Robotics: ROS 2 (Ubuntu), PID Control, Motion Control, Embedded Systems, Arduino, Sensor Interfacing
AI & Computer Vision: Google ADK, Gemini API, FastAPI, Prompt Engineering, Image Processing, Object Detection, AI-based Visual Recognition
CAD, Simulation & EDA: SolidWorks, MATLAB & Simulink (Intermediate), Simscape, KiCad, Proteus

CERTIFICATIONS

- Automation in Manufacturing - IIT Guwahati, NPTEL (Elite, 67%)
- Foundations of Prompt Engineering - AWS Training and Certifications

LEADERSHIP & ACTIVITIES

- Placement Representative (2026-Present) - Coordinate communication between students and placement coordinators.
- Member, ARME (2024-Present) - Association of Robotics and Mechatronics Engineers (2024-Present).
- YRC Volunteer, MIT (2024-2026) - Organized Youth Fest events for visually challenged individuals.

LANGUAGES

English (Proficient) | Tamil (Proficient) | Telugu (Conversational) | Hindi (Elementary)